

La Casita

My Disney-Inspired Smart & Sustainable Home with AI

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Powered by

AI, Imagination and Green Ideas!

Purpose of This Project

Learn Sustainability

To learn about sustainability and how homes can be designed to protect our planet.

Explore Technology

See how AI and tech can make homes smarter, greener, and more fun!

Building of the Future

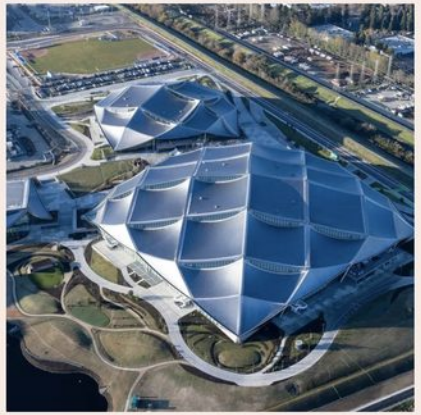
Create homes that help the planet, not harm it.

What is Sustainability?

Sustainability means meeting our needs without harming the planet.

What is a Sustainable Home?

A sustainable home is built and operated in ways that reduce the owner's carbon footprint and minimize impact on the environment and climate.



Google's Office



Amazon's Office



Apple's Office

My Question & Hypothesis

My Question

Are our homes Sustainable like Google's and Apple's offices? Or are we hurting the planet?

My Hypothesis

Most homes are already great just the way they are, but making a few small changes can help make them more sustainable.

My Prediction

Even if homes try to save energy, some appliances still use a lot. By testing, I can find better ways to use less power and help the planet.



Procedure

1

Pick the Big Appliances

I chose 8 big appliances in our home: Laundry Dryer, AC, Oven, LED Lights, Water Heater, Tesla Charger, Refrigerator, and Dishwasher.

2

Think of Better Ways

For each appliance, I researched if there's a more *sustainable* or eco-friendly option (like drying clothes in the sun instead of using a dryer).

3

Plan the Test

I made a chart to track electricity use. I tested each appliance on **2 different times** between **Dec 2024 and Feb 2025**.

4

Measure the Energy

Before I turned on the appliance, I wrote down the **electric meter reading**. I used the gadget for **1 hour**, then wrote down the **end reading**.

5

Repeat for Accuracy

I repeated the test **2 times** for each appliance on **different dates** to get good average results.

6

Do the Math

I calculated how much electricity each appliance used in an hour, how much it would use in a **month**, and how much it would **cost in money and carbon**.

7

Compare and Decide

I looked at which appliance use the most energy and which ones have better sustainable choices.

Data Collection

Collected energy used by 8 appliances in one hour. Energy usage is calculated as: End Reading - Start Reading.

Table 1: Individual Appliance Experiments (Meter Readings & Dates)

Appliance	Date	Start Meter (kWh)	End Meter (kWh)	Energy Used (kWh)
Laundry Dryer	Dec 07, 2024	5102	5111	9
	Dec 09, 2024	5340	5349	9
AC	Dec 21, 2025	6032	6035	3
	Dec 23, 2025	6360	6363	3
Oven	Jan 06, 2025	7100	7107	7
	Jan 08, 2025	7451	7458	7
LED Lights	Jan 18, 2025	8001	8002	1
	Jan 22, 2025	8321	8322	1

Appliance	Date	Start Meter (kWh)	End Meter (kWh)	Energy Used (kWh)
Water Heater	Jan 27, 2025	9002	9005	3
	Jan 29, 2025	9365	9368	3
Tesla Charger	Feb 01, 2025	10025	10032	7
	Feb 03, 2025	10181	10188	7
Refrigerator	Feb 06, 2025	11000	11001	1
	Feb 10, 2025	11374	11375	1
Dishwasher	Feb 15, 2025	12010	12011	1
	Feb 17, 2025	12365	12366	1

kWh = Kilo Watt Hour is the unit of measurement of electricity

Data Processing

Assuming average household monthly usage, and using approximate costs: $0.15/kWh electricity & 0.10 carbon cost.

Table 2: Average Monthly Usage and Cost Estimates

Appliance	Rounded Avg Energy Use (kWh)	Typical Use (hrs/month)	Energy Used (kWh/month)	Rounded Bill ($0.15/kWh)	Rounded Carbon Cost ($0.10/kWh)	Total Monthly Cost
Laundry Dryer	9	15	135	$20	$14	$34
AC	3	50	150	$23	$15	$38
Oven	7	20	140	$21	$14	$35
LED Lights	1	30	30	$5	$3	$8
Water Heater	3	40	120	$18	$12	$30
Tesla Charger	7	30	210	$32	$21	$53
Refrigerator	1	100	100	$15	$10	$25
Dishwasher	1	60	60	$9	$6	$15

Total Bill: $143, Total Carbon Cost: $95, Total Monthly Cost: $238

kWh = Kilo Watt Hour is the unit of measurement of electricity

My Testing Method

To make sure my results were fair and accurate, I followed the scientific method.

Controls

These are the parts of the experiment that I did not change

- Each appliance was tested for **1 hour**
- T**urned off** all the other appliances during testing
- Tested each appliance **twice**
- The **same house** was used for all tests
- I used each appliance the **same way every time**

Variables

These are the parts that changed

Independent Variables (What I changed)

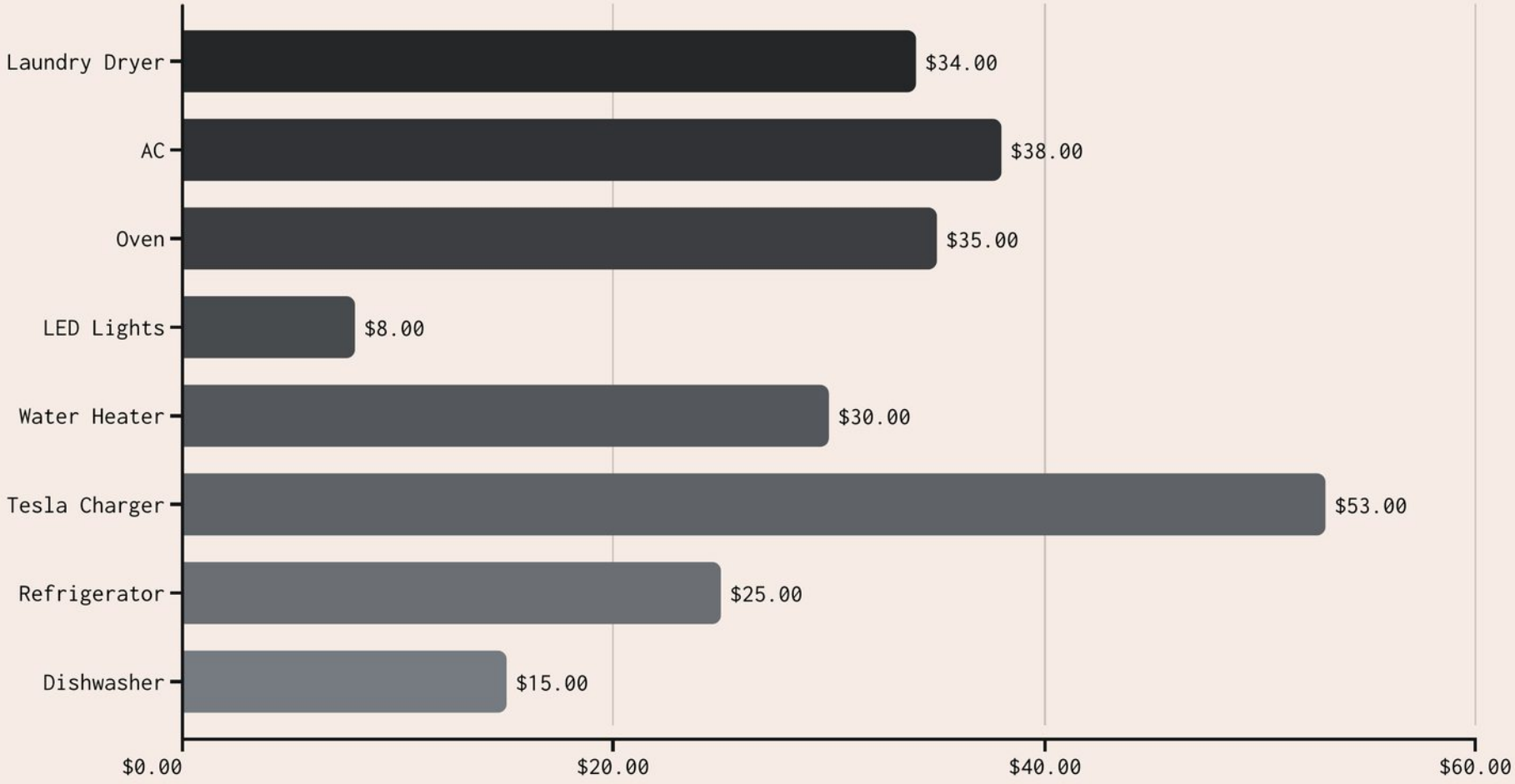
- The type of appliance (dryer, AC, oven, etc.)
- The day of testing
- Size of appliance (AC vs LED lights)
- Function of the appliance (Heating, Lighting, Cooking, etc.)

Dependent Variables (What I measured)

- How much energy (kWh) each appliance used in 1 hour
- Estimated monthly energy use
- Electricity cost per month
- Carbon footprint (how much pollution was created)

Data Analysis

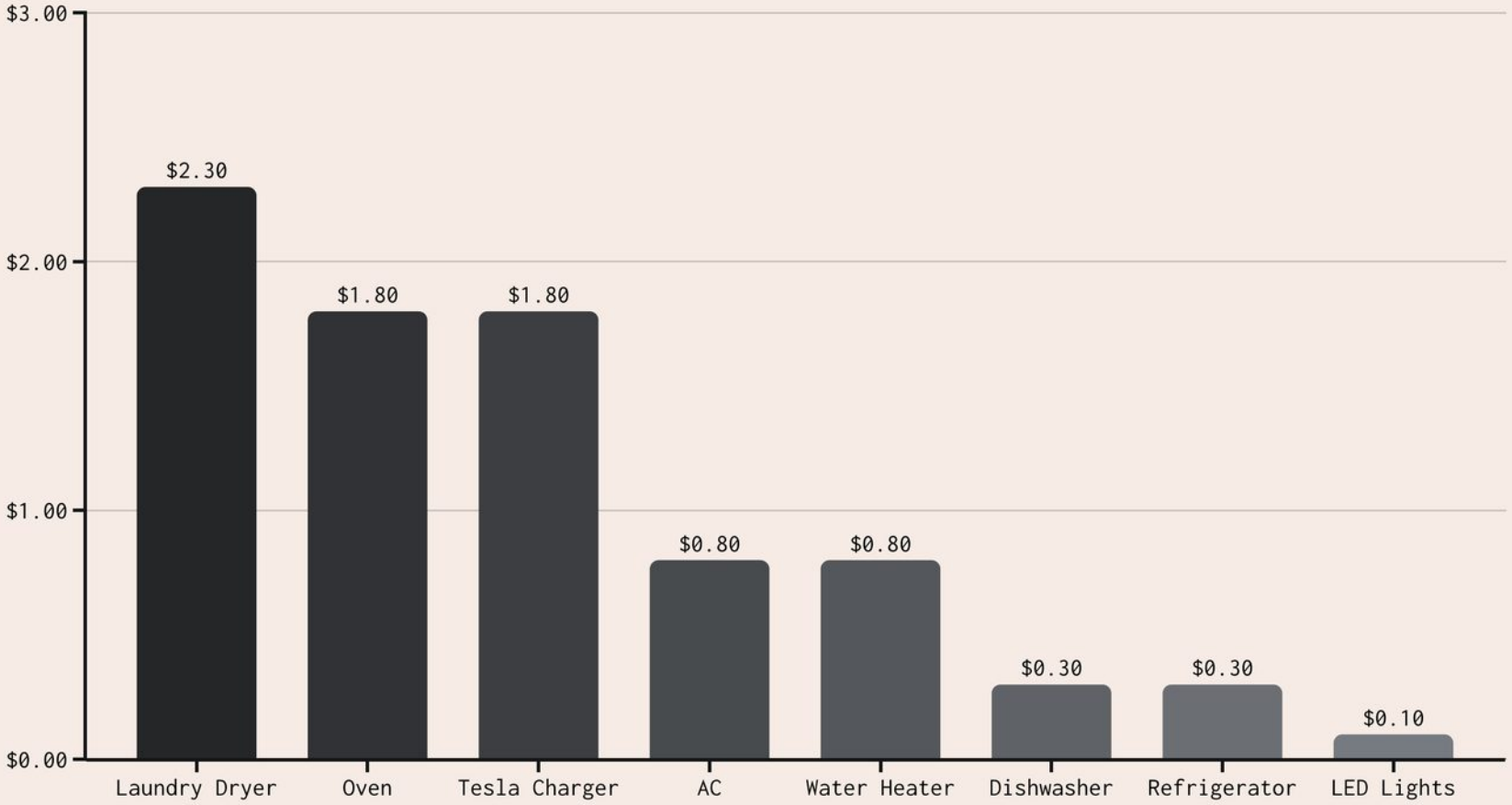
Chart 1: Total Monthly Cost by Appliance



Total Bill: $143, Total Carbon Cost: $95, Total Monthly Cost: $238

Data Analysis

Chart 2: Cost per Hour by Appliance

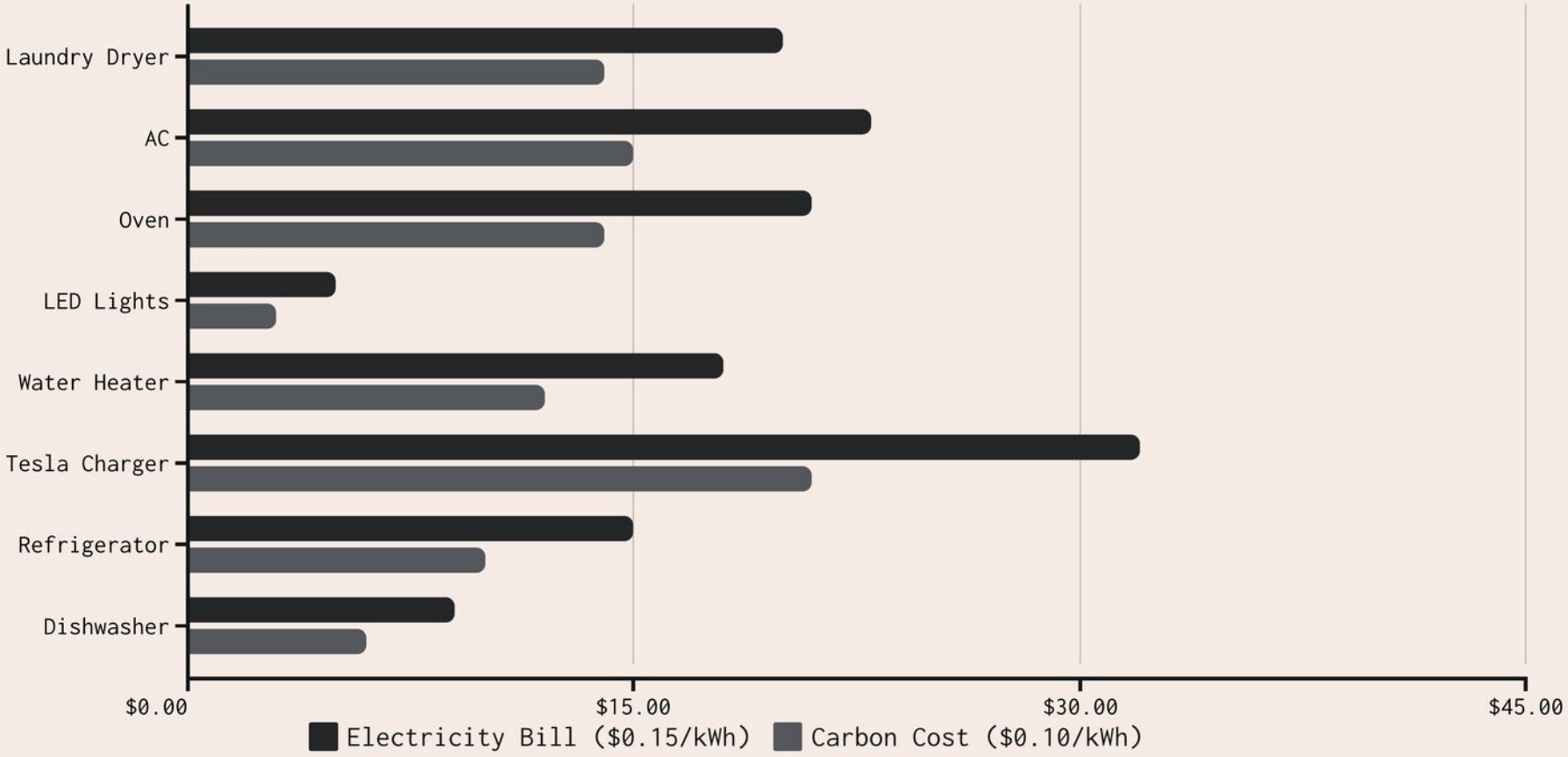


'Laundry Dryer' & 'Oven' have a high cost per hour of use.

Cost per hour of Use = (Total Monthly Cost, $) / (Typical Use (hrs/month)'

Data Analysis

Chart 3: Comparison of Electricity Bill & Carbon Cost



Carbon Cost is pretty significant.

Key Takeaways

'Tesla Charger' has highest $ cost.

Optimize Tesla charging schedule.

'Laundry Dryer' & 'Oven' have a high $ cost per hour of use.

Explore alternative sustainable methods to save money.

'LED Lights' have lowest energy use.

Very Good.

Carbon cost is as high as $ cost.

Conclusion: What did I learn?

Appliance	Sustainable Alternative	Can Replace?
Laundry Dryer	Sun-drying clothes	✅ Yes
Air Conditioner	Ceiling fan or open windows	✅ Yes
Big Oven	Solar cooker	❌ No (not always)
LED Lights	Already efficient	✅ Keep
Water Heater	Solar water heater	❌ No (hard to install)
Tesla Charger	Charge during solar hours	✅ Yes
Refrigerator	Energy Star fridge	✅ Yes
Dishwasher	Hand washing (with less water)	✅ Yes
Pool Heater	Solar pool heater	✅ Yes

Part II

Explore Technology

See how AI and technology can be used to find practical alternatives!

Designing La Casita

This is where design thinking, engineering, and fun all came together!

Inspiration

Inspired by Google's office and Disney's magical "Casita" from *Encanto*, I created a smart home that's full of **technology and heart**.

Tools Used

AI Named Mirabel

Runs everything behind the scenes with magic!

Raspberry Pi Pico W

Controls all the lights in the home.

iPhone App

Sends voice commands to the house.

Face Recognition

Opens doors and starts conversations.



How Mirabel Works

- Mirabel is a fun, Disney-themed AI home assistant
- She understands voice commands and manages all home devices

Voice Command

I talk to the **iPhone app**

AI Processing

It sends the request to **Mirabel's server**

Smart Response

Mirabel understands what I want and tells the house what to do

This helped me learn how AI connects to real devices. AI Coding feels like magic!



Future Plans : Helping Our Planet with Green Tech

La Casita is **eco-friendly and an ever-growing** project! I'll keep adding green solutions every year.



Hydroponics

Growing plants without soil, saving **90% of water**



Clay fridge

Keeps water/food cool without any electricity



Atmospheric Water Generator

Pulls clean drinking water from the air *(up to 10L/day)*



Free mesh internet

Community-powered access



Composting

Turns food waste into plant food *(reduces waste by up to 50%)*



Solar power

To use sunlight instead of electricity *(can save 2,000 lbs of CO_2 per year!)*



Harvest Rainwater

Collect rainwater to save water and help nature.



No AC

Cross ventilated homes and Trees eliminates AC

In Conclusion: What I Learned

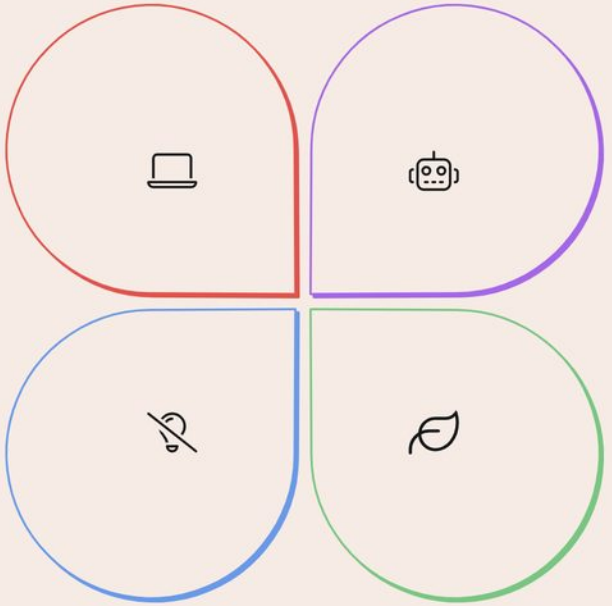
La Casita is my way of showing the future can be magical AND sustainable.

Tech Skills

How to connect hardware and software

AI Knowledge

What AI really is and how to use it kindly



Big Discovery

Even a 3rd grader can help change the world!

Green Science

Real-world climate science and eco-technology

> "Even a small idea can make a big impact – and so can a 3rd grader!"



Thank You!

I worked on this project **with my parents**. We researched, designed, learned, coded, tested, and laughed together. My Dad and AI helped with the hard tech stuff – I helped with ideas, design, testing, and storytelling! 💛

AI Magic

Smart technology

Sustainability

Earth-friendly solutions

Disney Inspiration

Creative design

Love & Learning

Growing together

🎉 Thank you for visiting La Casita! I hope you enjoyed the magic of AI, Disney, and sustainability all in one home.

Have questions? Mirabel and I are ready! *(She may even tell you a joke... if she's in the mood!* 😉*)*